

|   |                                                                                                         |          |
|---|---------------------------------------------------------------------------------------------------------|----------|
| 5 | (a) (i) movement/flow of charge carriers                                                                | B1       |
|   | (ii) $\frac{\text{work (done) or energy (transformed)(from electrical to other forms)}}{\text{charge}}$ | B1       |
|   | (b) (i) p.d. across one lamp = 2.5 V<br>resistance = $[(8.7 - 7.5)/0.3]/2 = 2.0 (\Omega)$               | C1<br>A1 |
|   | (ii) straight line through the origin<br>with gradient of 0.5                                           | M1<br>A1 |

| Page 4 | Mark Scheme                                     | Syllabus | Paper |
|--------|-------------------------------------------------|----------|-------|
|        | Cambridge International AS/A Level – March 2016 | 9702     | 22    |

- (iii)  $P = I^2 R$       or  $P = VI$  and  $V = IR$       or  $P = V^2 / R$  and  $V = IR$       C1  
 $= 0.30^2 \times 2.0$        $= 0.60 \times 0.30$        $= 0.60^2 / 2.0$   
 $= 0.18 \text{ (W)}$       A1
- (iv) 1  $R = \rho l / A$       C1  
 $l = (2.0 \times 0.40 \times 10^{-6}) / 1.7 \times 10^{-8}$   
 $= 47 \text{ (m)}$       A1
- 2  $I = Anvq$   
 $v = 0.30 / (0.40 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$       C1  
 $= 5.5 \times 10^{-5} \text{ (m s}^{-1}\text{)}$       A1